

AI-POWERED

Credit Risk Intelligence Platform

Not just a model — a credit decision system: scored, explained, defensible, auditable, and queryable in plain English.

Samiksha Batra

AI Engineer — Candidate Assignment

Live demo <https://ai-powered-credit-risk-intelligence-platform-8eftb3rvzqybqg2pj.streamlit.app/>

Repository <https://github.com/Samikshabatra/AI-Powered-Credit-Risk-Intelligence-Platform>

Faster credit decisions the bank can defend

Lenders must decide quickly and accurately — and, increasingly, explain and audit every decision. The platform targets six real needs in one workflow.

1 Spot risk early

Flag high-risk applicants before approval

2 Automate scoring

Straight-through decisions with a risk band

3 Explain decisions

Principal reasons for every classification

4 Satisfy audit

Traceable, regulator-ready decision path

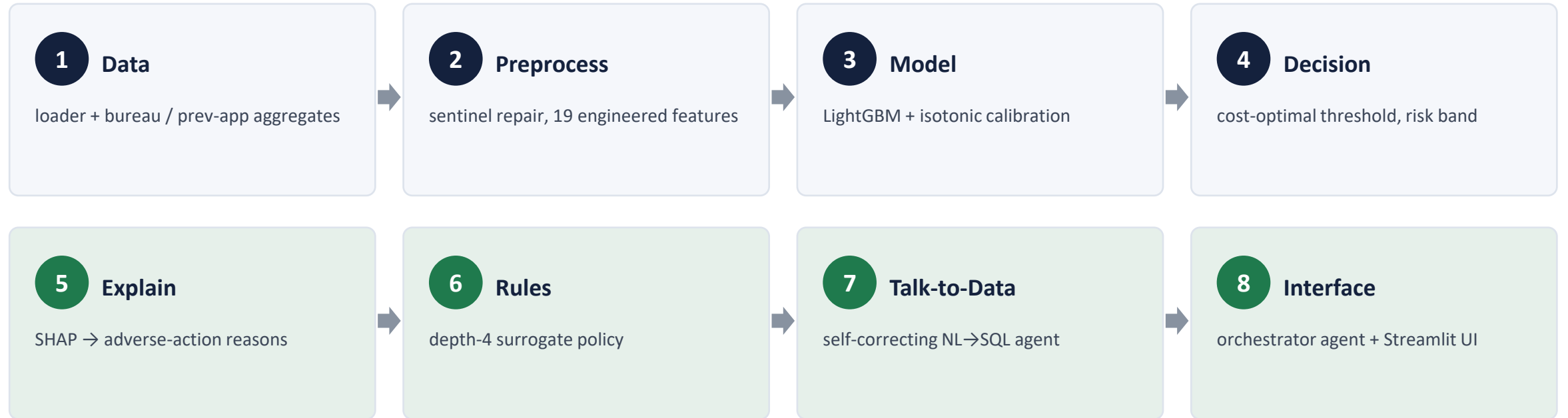
5 Analysts in plain English

Query the portfolio without SQL

6 Bridge ML & policy

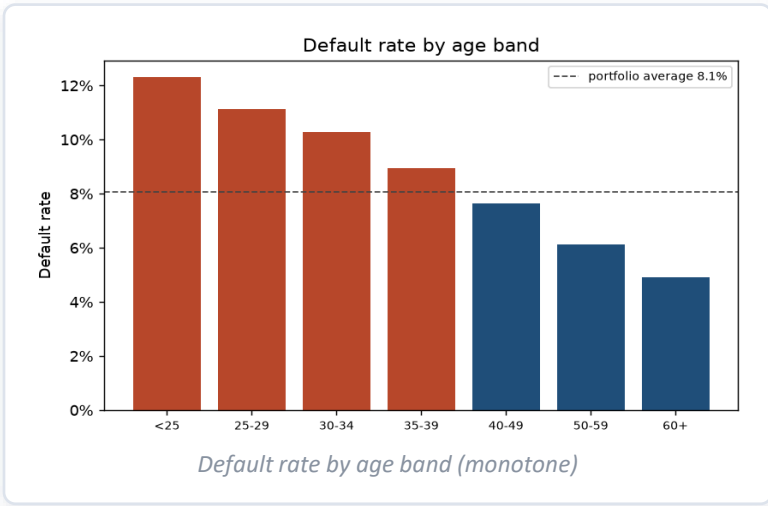
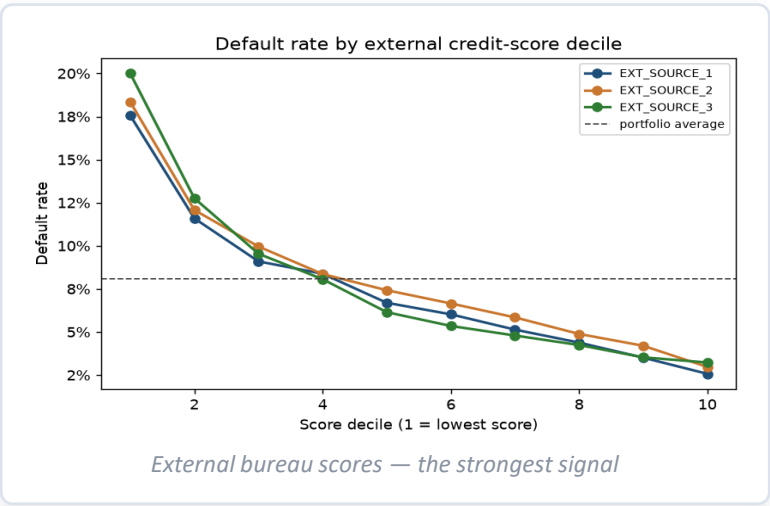
Model insight expressed as credit rules

One pipeline, end to end



Runs two ways from one image — docker compose up for the local grader, and a deployed public URL. SQLite is read-only at query time.

Five insights, and one that changes the model



The finding worth reading

DAYS_EMPLOYED = 365243 on 18% of rows is a “not employed” sentinel, not a 1000-year career.

99.96% are pensioners — who default at 5.4%, below the 8.7% employed average.

Repaired in the preprocessor, the semantic layer and the EDA, so it can never leak back.

307,511
applications

8.07%
default base rate

143
features (19 engineered)

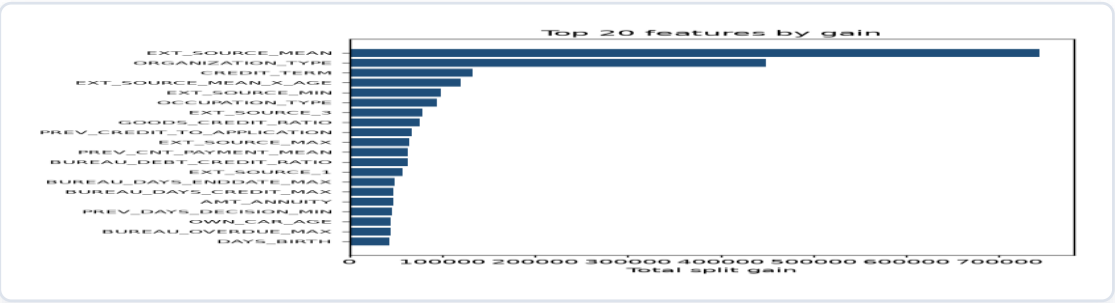
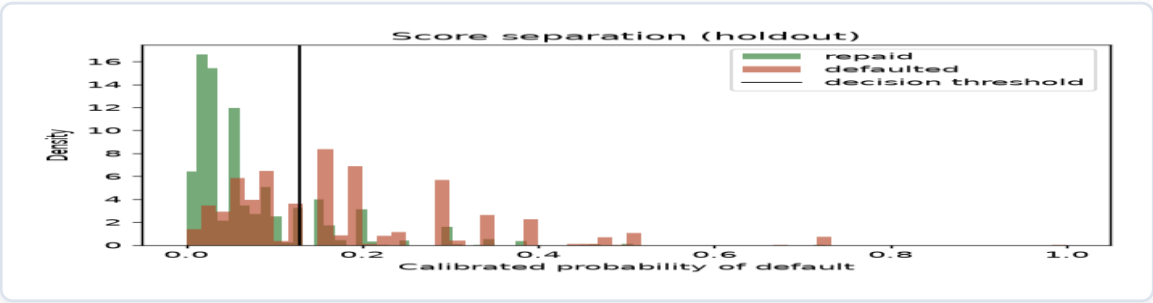
18%
sentinel rows repaired

An honest pipeline built for imbalance

- LightGBM — native NaN handling (missingness is signal) and native categoricals (no 200-column one-hot); SHAP-friendly.
- Class imbalance via `scale_pos_weight = 11.39`, not SMOTE — synthesising minority rows across categorical codes invents applicants that cannot exist.
- Four disjoint stratified splits so no number is reported on data that shaped it.
- Missing values kept as NaN — “no bureau record” and “no external score” are themselves predictive.

Split	Share	Used for
Train	60%	Fit the booster
Valid	10%	Early stopping
Calib	10%	Calibrator + threshold
Holdout	20%	Every headline number

scale_pos_weight 11.39 · 1,190 boosted trees · isotonic calibration



MODEL PERFORMANCE

Measured on 61,503 unseen applicants — and versus baselines

0.7788

Holdout ROC-AUC

0.2637

PR-AUC (8% base rate)

0.415

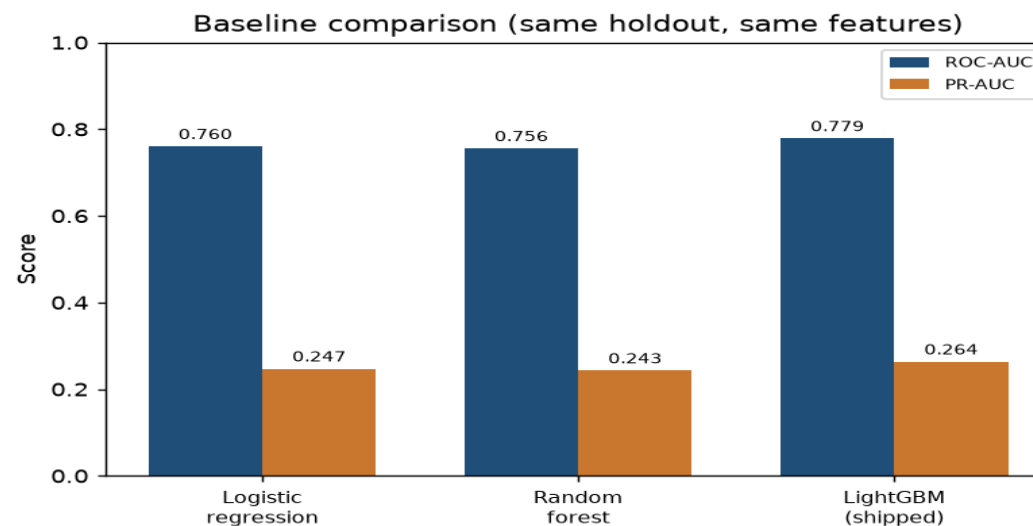
KS statistic

3.66×

Top-decile lift

Model	ROC-AUC	PR-AUC	KS
Logistic regression (balanced)	0.7605	0.2474	0.389
Random forest (balanced)	0.7557	0.2430	0.380
LightGBM (shipped)	0.7788	0.2637	0.415

Same holdout, same features, same seed — LightGBM wins both metrics; the linear model reaching 0.76 shows the signal is largely additive.



Holdout ROC-AUC & PR-AUC by model

Probabilities you can price a portfolio on

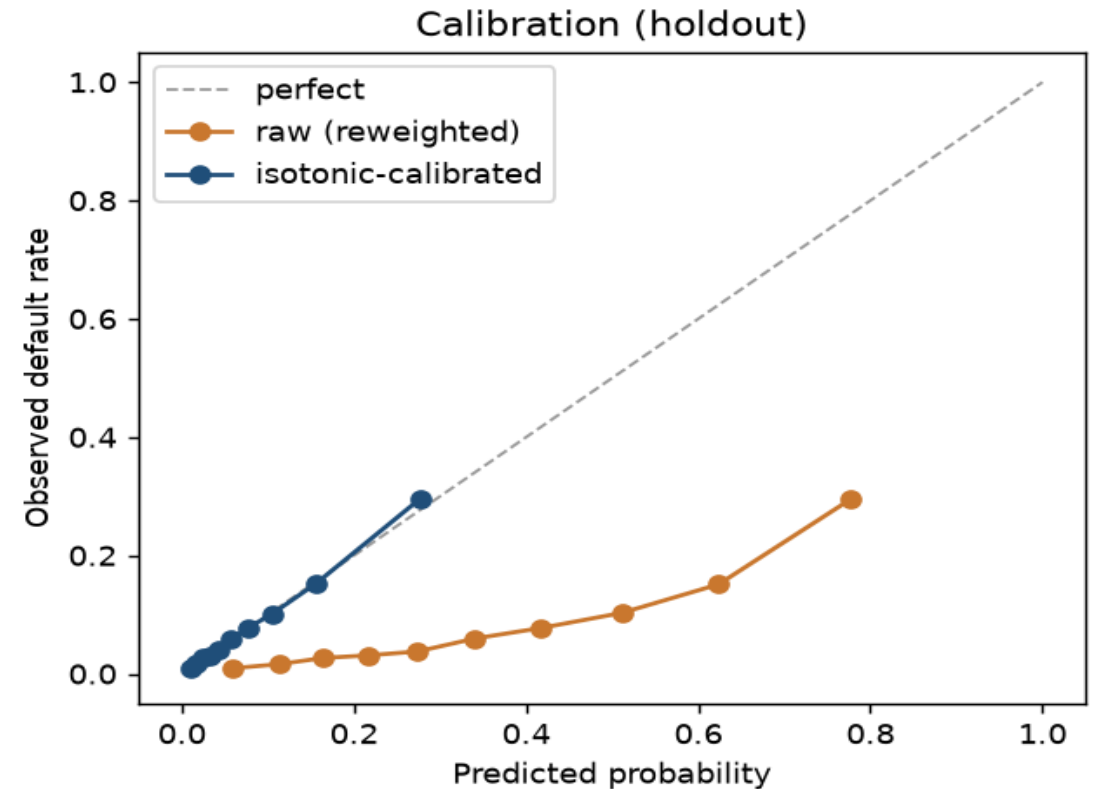
- Imbalance reweighting fixes the ranking but wrecks the probabilities — the raw model predicts ~46% average risk against an actual 8.1%.
- Isotonic regression, fitted on a split used for nothing else, restores them without reordering a single applicant.
- Now “0.30” means 30 defaults per 100 — the number a lender prices against.

0.160 → 0.066

Brier score

0.269 → 0.004

Expected calibration error

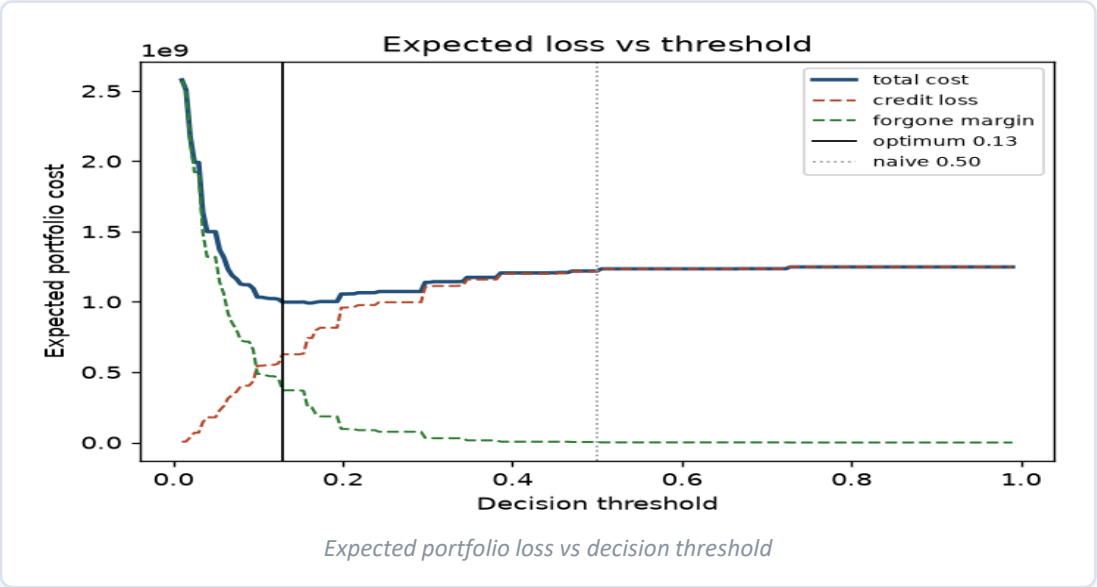


Reliability curve — raw vs isotonic-calibrated (holdout)

The threshold is 0.129, not 0.50

- A false negative costs 45% of the loan (loss given default); a false positive 8% (forgone margin) — each weighted by that applicant’s exposure.
- The threshold that minimises expected loss is 0.129, approving 81.4% of applicants.

Policy	Expected loss	Saving
Approve everyone	\$1.25 B	\$250 M
Naive 0.50 threshold	\$1.22 B	\$221 M
Cost-optimal 0.129	\$998 M	—



Risk bands anchored on economics: Low < 6.4% Medium 6.4–12.9% High ≥ 12.9%

Every decline gets its principal reasons

- SHAP TreeExplainer — exact for tree ensembles, milliseconds per applicant.
- Adverse-action reason codes under ECOA / Reg-B: SHAP contributions become the plain-English reasons a lender must legally give a declined applicant.
- Two layers: deterministic templates first, LLM as a stylist over those facts — it cannot invent a reason the model did not produce.
- Protected attributes are suppressed from the notice and reported separately.

SAMPLE ADVERSE-ACTION NOTICE

Decision: Decline

High risk — 38.3% probability of default (threshold 12.9%).

Principal reasons

1. External credit scores are below the approved-applicant average.
2. The implied repayment term is longer than typical.
3. A high share of previous applications were refused.

Gender and age were excluded from these reasons by policy.

The model, expressed as credit policy

A 1,190-tree ensemble is not policy. A depth-4 tree fitted to the model’s own decisions is — its leaves read back as IF-THEN rules, each with support, precision and lift.

Rule (IF ... THEN decline)	Support	Default rate	Lift
Avg external score ≤ 0.26 AND weakest ≤ 0.18	5.1%	28.2%	3.50×
$0.26 < \text{avg score} \leq 0.36$ AND weakest ≤ 0.18	4.5%	18.7%	2.33×
Avg score ≤ 0.36 AND weakest > 0.18 AND low down-payment cover	3.6%	16.8%	2.09×

14

IF-THEN rules

81.2%

Surrogate fidelity

Numeric only

no encoding artefacts

No protected

attributes in rules

Ask in plain English, get SQL-backed answers

Q. Which 5 occupations have the highest default rate (≥ 1000 applicants)?

```
SELECT OCCUPATION_TYPE, AVG(TARGET) AS default_rate, COUNT(*)  
  
FROM applications  
  
WHERE OCCUPATION_TYPE IS NOT NULL  
  
GROUP BY OCCUPATION_TYPE  
  
HAVING COUNT(*) >= 1000  
  
ORDER BY default_rate DESC LIMIT 5;
```

A. Low-skill labourers lead at 17.2%, followed by drivers (11.3%) and waiters/barmen (11.3%).

- Self-correcting agent — on a SQL error the message goes back to the model, which rewrites the query.
- Semantic layer pins canonical metric definitions so the model never re-invents “default rate”.
- Four guardrail layers: SELECT-only allowlist, keyword denylist, table allowlist, read-only connection.

NL→SQL eval (25 cases)	
Execution accuracy	100%
Answer accuracy	90.5%
Fallback / injection caught	100%
Retry rate	0%

The Decision Trace — every module in one screen

One chat surface routes to six tools. For any applicant the whole chain renders top to bottom — the platform’s answer to “auditable”.

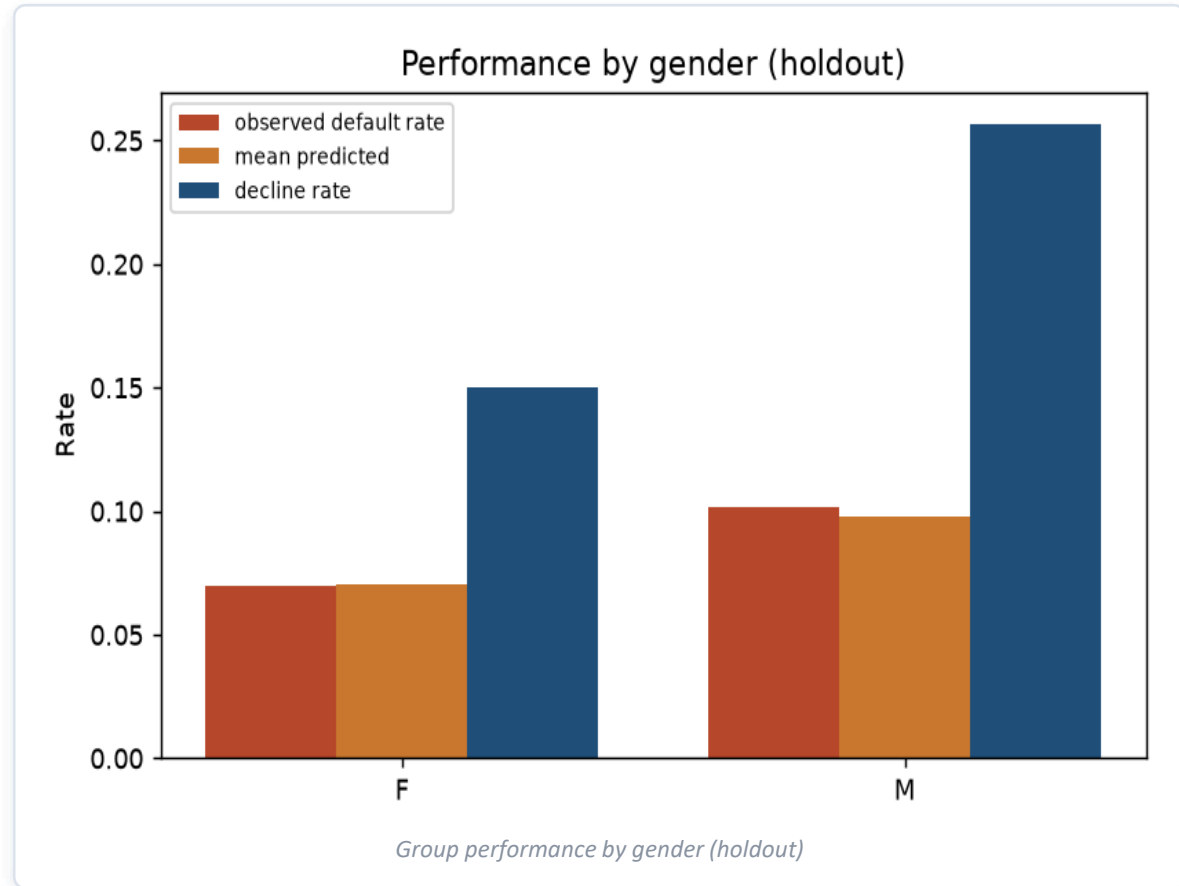
1	Applicant #170169	143 features · 19 engineered · sentinel repaired
2	Risk model	raw score 0.859 (LightGBM, 1,190 trees)
3	Calibration	38.3% probability of default
4	Decision threshold	decline at or above 12.9% (cost-optimal)
5	Risk band	HIGH
6	Explanation	low external scores · long term · 75% prior refusals
7	Policy rule	avg score $\leq 0.26 \rightarrow$ DECLINE (lift 3.50×)

DECISION

DECLINE

Reduced limit
+ collateral only

Governance measured, not hidden



- Approval-rate gap of 10.7pp by gender and 30.5pp by age band — published on screen, not silently absorbed.
- Which parity to equalise is a legal and policy choice, so the platform measures rather than quietly “fixes” it.
- Distribution shift is labelled honestly: the dataset has no time axis, so it is a split check, not fake “drift”.

10.7 pp

Gender approval gap

30.5 pp

Age-band approval gap

Optimised, guarded, tested, containerised

60.6%

Input-token saving (prompt caching)

124

Automated tests passing

4-layer

SQL guardrails

Multi-stage

Docker + compose

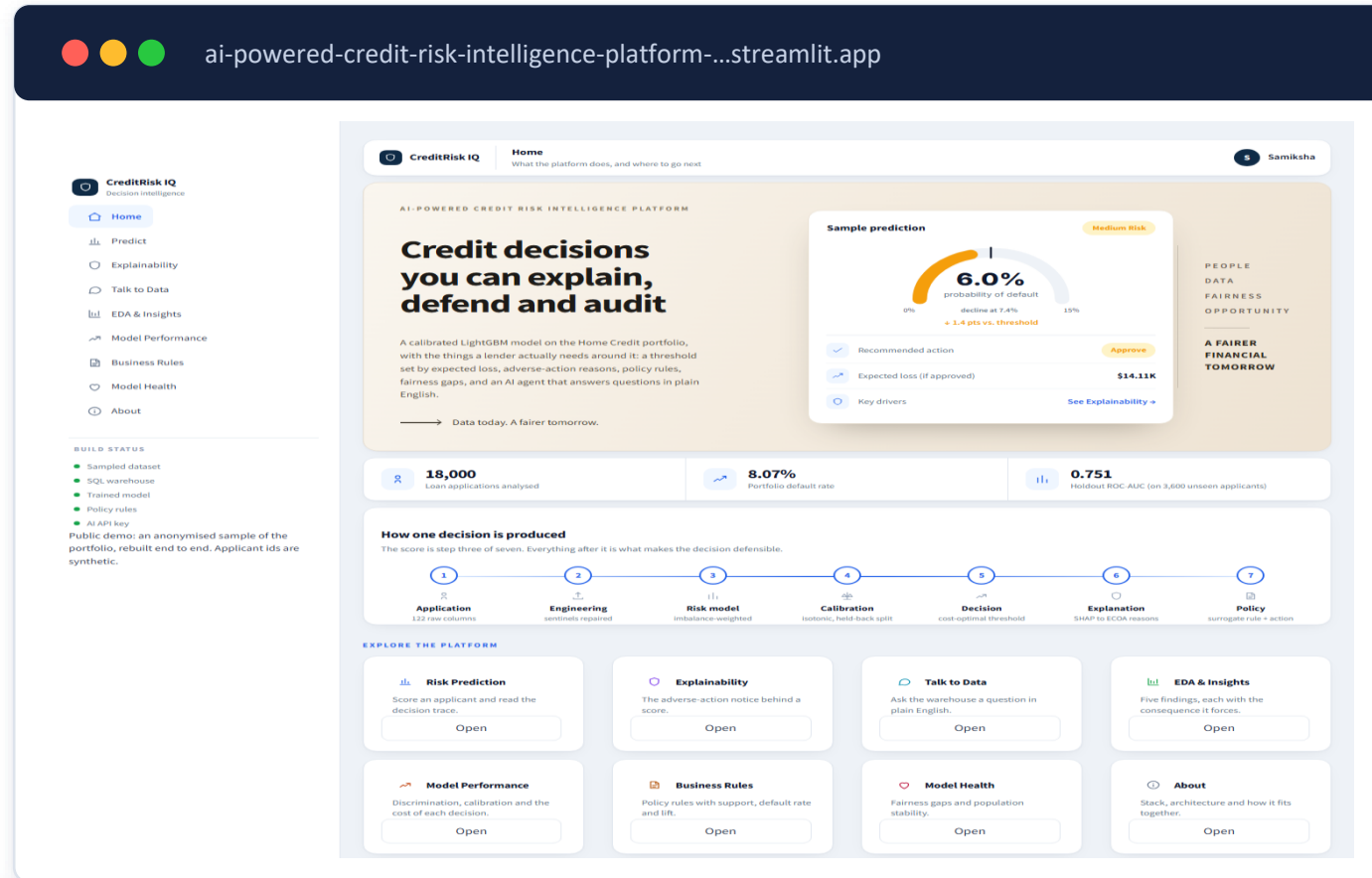
Prompt & token optimisation

- Stable prefix (schema + metrics + exemplars) cached with `cache_control` — 72.3% cache-hit rate on the live run.
- Cheap model summarises result rows; the strong model's context never grows by them.
- Hallucination control: schema-grounding, semantic layer, and a hard `CANNOT_ANSWER` fallback.

Reliability & delivery

- Tests span guardrails, semantic-layer SQL run against the live database, and compliance checks (no protected attribute ever surfaces as a reason).
- Multi-stage build, non-root user, health-check; docker compose up runs the whole app.
- Artifacts and dataset are git-ignored — never committed.

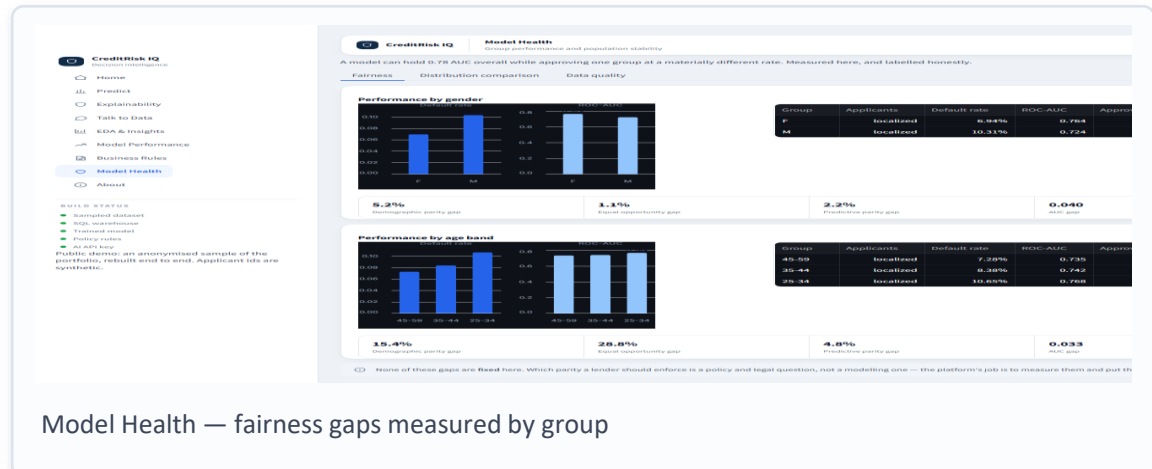
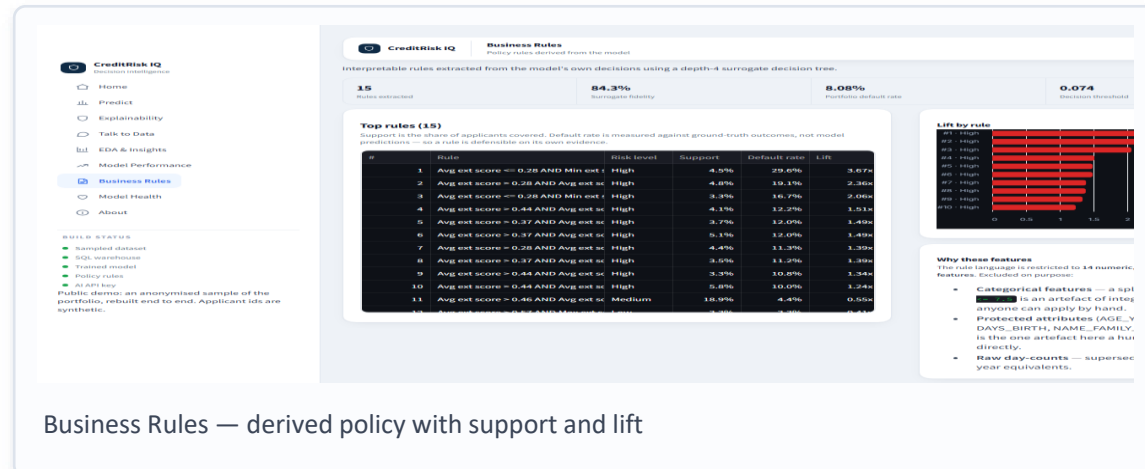
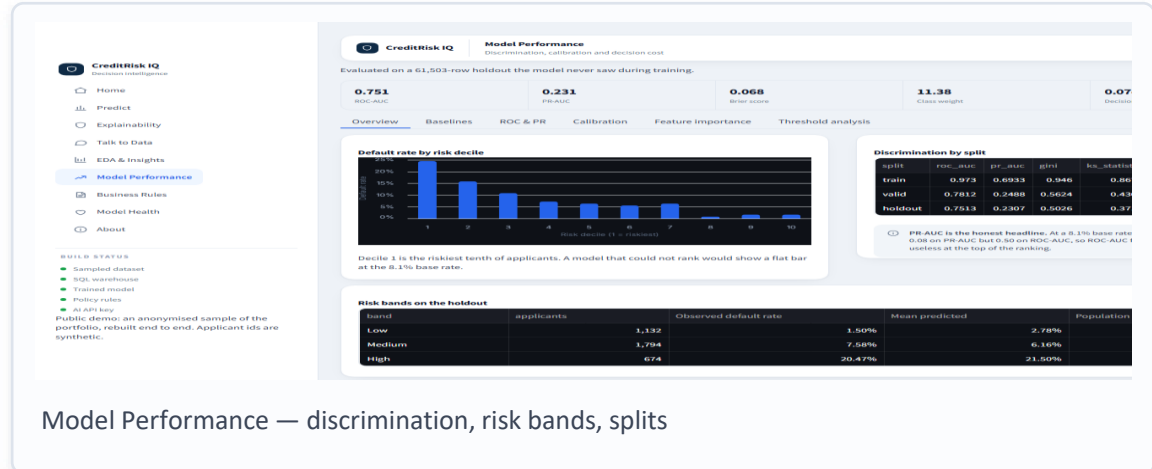
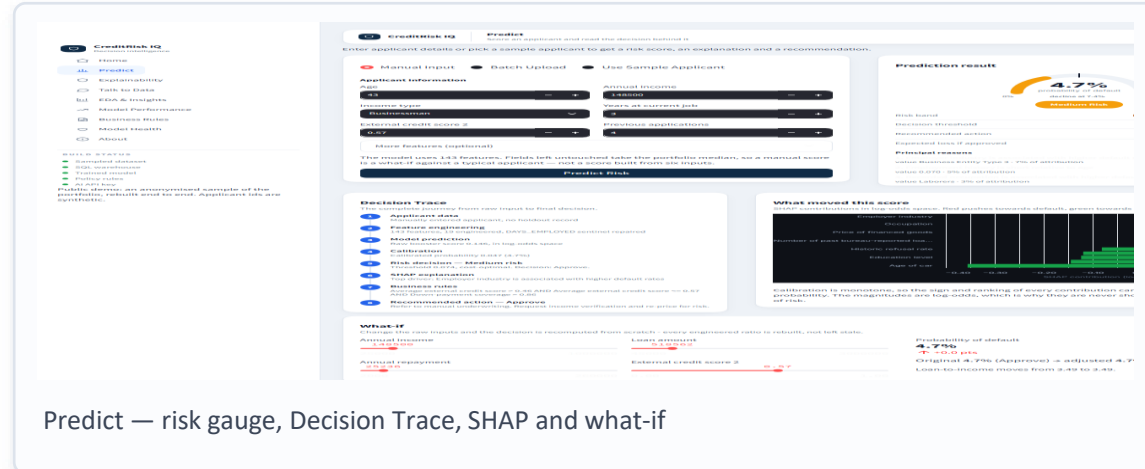
One image, two ways to run



- Local grader: docker compose up runs the full app from one command.
- Public link: deployed on Streamlit Community Cloud, live and shareable.
- The API key is injected at runtime as a host secret — never in the image or git.
- Public demo runs an anonymised 18k-row sample (ROC-AUC 0.751); the full model scores 0.7788.

LIVE DEMO

<https://ai-powered-credit-risk-intelligence-platform-8eftb3rvzqybqg2pj.streamlit.app/>



Results, limitations and what comes next

What was delivered

- Calibrated 0.7788-AUC model
- Cost-optimal threshold — \$221M saved vs 0.50
- SHAP adverse-action reason codes
- 14 auditable policy rules
- Self-correcting NL→SQL agent (100% execution)
- Fairness measured and published
- Dockerised and deployed live

Known limitations

- No temporal validation — one snapshot, no time axis.
- Cost matrix (LGD 45% / margin 8%) is assumed, not sourced.
- Fairness is measured, not mitigated.
- Reason codes need a human / counsel review loop.
- AUC ceiling ~0.78 on the three tables used.

Future work

- Add installments-payment features.
- Per-segment thresholds (cash vs revolving).
- Reject-inference for survivorship bias.
- A real drift monitor once time-ordered data exists.

Thank you

A credit decision system a bank could actually deploy — scored, explained, defensible, auditable.

Samiksha Batra · AI Engineer — NeoStats Assignment

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